

Pranav Shankar

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EDUCATION

Cornell University, College of Engineering, Ithaca, New York **Expected December 2026**
Bachelor of Science in Mechanical Engineering, Minor in Aerospace Engineering **Cumulative GPA: 4.10**
Courses: Fluid Mechanics, Thermodynamics, System Dynamics, Intro to Aeronautics, Spaceflight Mechanics

TECHNICAL EXPERIENCE

Cornell Rocketry Team Ithaca, New York
Liquid Engine Static Fire Stand Responsible Engineer July 2025–Present

- Leading structural design and analysis of the static fire testbed for the newly developing liquid propulsion engine
- Collaborating with propulsion subteam to safely and optimally integrate piping, instrumentation, and electronics
- Presenting analysis-based designs and design intent in formal design reviews and written technical documents

Fin Redesign Responsible Engineer August 2024–June 2025

- Designed airfoiled fins and aerodynamic fin mounts based on trade study analysis and stability requirements
- Verified structural integrity under vibrations and flight loads using quasi-static Ansys FEA and MATLAB analysis
- Improved carbon fiber fins' strength and production via new heat treatment process, aiding third place IREC win

Rapid Component Design Structures Engineer December 2023–June 2024

- Engineered 3D printed fin mount additions using SOLIDWORKS that reduced over 40% drag on the fin assembly
- Designed and integrated a GPS system harness with pitot tube avionics and hardware, ensuring secure placement with minimal electronic/mechanical interference, error-proofing placement and simplifying installation

Polamer Precision, Inc. New Britain, Connecticut
Manufacturing Engineering Intern May 2025–August 2025

- Created an end-to-end manufacturing process for a military aircraft engine component; enhanced quality on 40+ manufacturing processes on four commercial engines using GD&T in SOLIDWORKS 3D modeling and 2D drafting
- Analyzed performance and modified a CNC macro, achieving 15% gain in efficiency and increased tool longevity
- Designed and analyzed a modular coolant nozzle, using SOLIDWORKS, MATLAB, and fluid mechanical principles
- Developed rework processes and determined MRB dispositions for over 120 non-conformances based on risks

ZT Group Cogeneration Project Ithaca, New York
Undergraduate Research Assistant January 2025–Present

- Researching phase-change materials for thermal energy management for use in solar cogeneration ovens
- Analyzing heat transfer data and preparing samples of phase-change material for later simulation-based analysis
- Characterizing system efficiency with steady-state and transient thermal finite element analysis (FEA) in Ansys

Cornell School of Mechanical and Aerospace Engineering Ithaca, New York
Undergraduate Teaching Assistant August 2024–Present

- Leading Mechanics of Engineering Materials labs on material testing (tension, fracture, etc.) per ASTM standards
- Taught discussions for Statics and Mechanics of Materials (Fall 2024) teaching static analysis and stability analysis

SKILLS & CERTIFICATIONS

Certifications: Cornell Engineering Leadership Certification (May 2025), SOLIDWORKS CAD Design Associate (July 2024)

CAD Design/Modeling Tools: SOLIDWORKS, OnShape, Autodesk Fusion, Siemens NX

Manufacturing: Geometric Dimensioning & Tolerancing (GD&T), carbon fiber composites, mill/lathe, 3D-printing

Software: Ansys FEA/FEM (Static Structural and Thermal), MS Office (Word, PowerPoint, Excel), Simulink

Programming Languages: Python, LaTeX, Embedded C/C++, MATLAB

LEADERSHIP & INVOLVEMENTS

American Institute of Aeronautics and Astronautics Cornell Branch, President **August 2025–Present**
American Society of Mechanical Engineers Cornell Section, Vice President **March 2024–Present**
Civil Air Patrol, U.S. Air Force Civilian Auxiliary, Cadet Officer **July 2021–September 2024**